机电系统建模、辨识与控制
Modeling, Identification and Control of Mechatronic Systems

MIC_L1 Modeling

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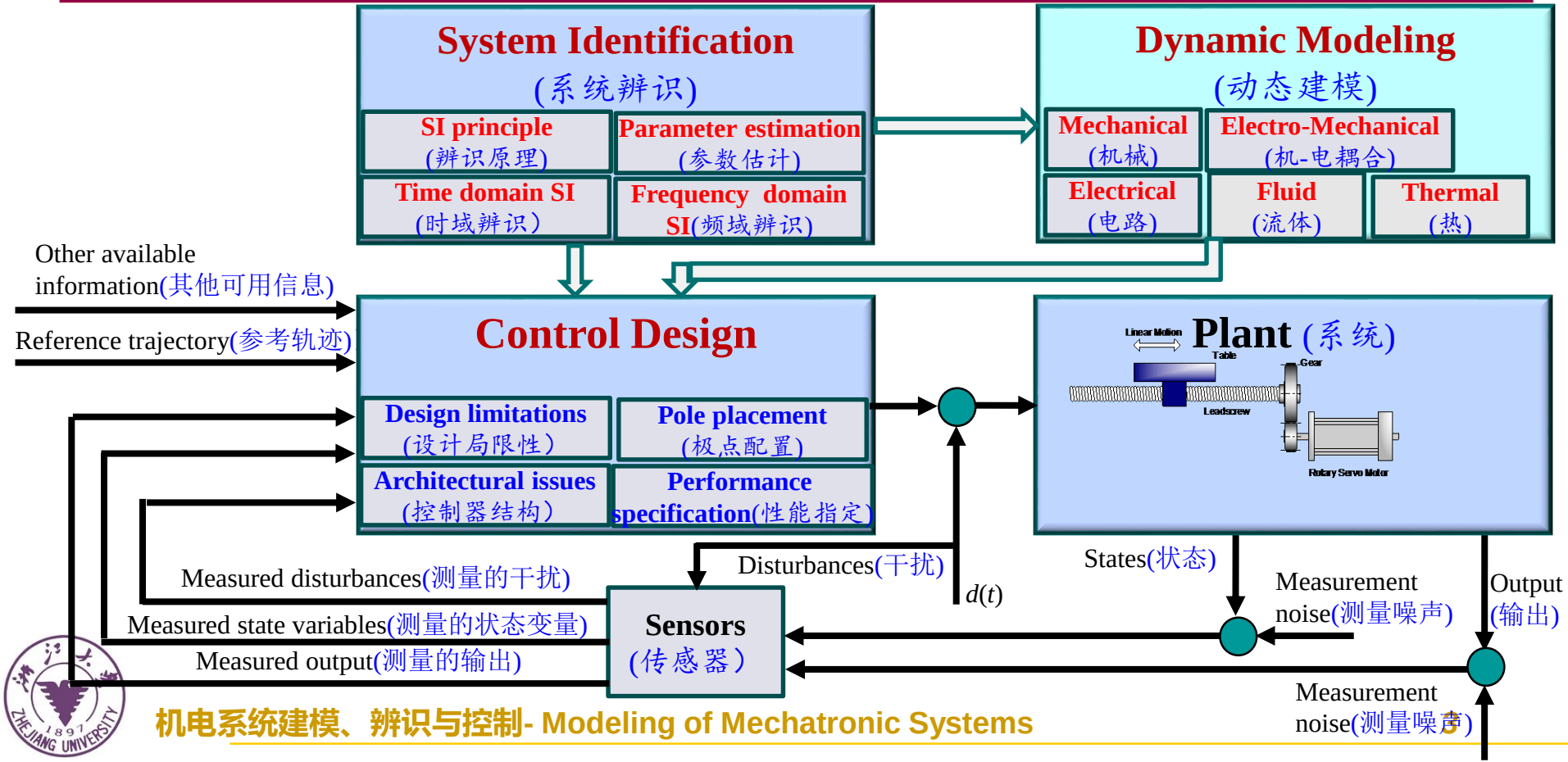
Modeling of Mechatronic Systems(机电系统建模)

- **Primary Elements** （主要元件）
- **Basic (Idealized) Modeling Elements** （基本(理想)建模元件）
- **Interconnection Relationships -Physical Laws** （互联关系–物理规律）
- **Energy Distribution** （能量分布）
- **Modeling Steps** （建模步骤）
- **Input/Output Models vs State-Space Models** （输入/输出模型和状态空间模型）

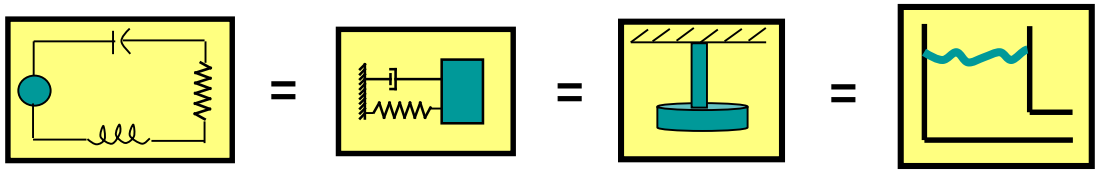
Main Contents of Course(课程内容)



Interdisciplinary and System Nature (跨领域与系统特性)

Analogous systems (相似系统)

- **Models are the same regardless of the physical domain of interest**
  (无论感兴趣的物理领域如何，模型都是相同的)

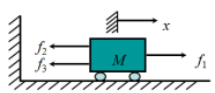
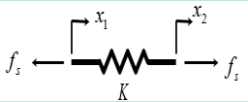
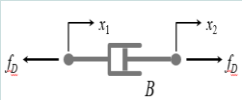
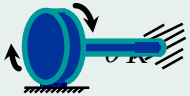
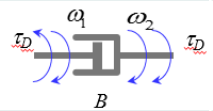
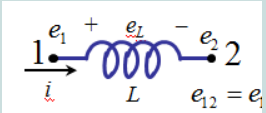
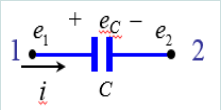
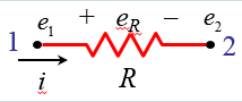


y = Charge	y = Transl. displ.	y = Angular displ.	y = Volume
u = Voltage	u = Force	u = Torque	u = Pressure

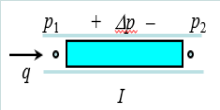
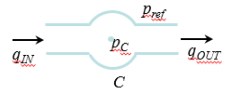
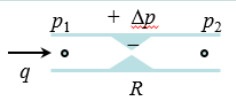
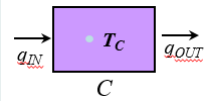
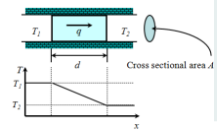
$$u \Rightarrow \ddot{y} + \dot{y} + y = u(t) \Rightarrow y$$

- ***We only need to understand how to analyze one model, but the results are applicable for four seemingly different types of physical systems!***
  (我们只需要了解如何分析一个模型，但结果适用于四种看似不同类型的物理系统）

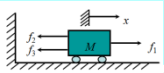
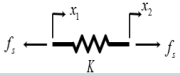
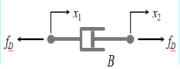
Primary Elements (主要元件)

System(系统)	Inertia Element: Store Kinetic Energy (惯性元件：储存动能)	Stiffness Element: Store Potential Energy (刚度元件：储存势能)	Friction Element: Dissipate Energy (摩擦元件：耗散能量)	Displacement variable(位移变量)	Velocity variable(速度变量)	Force variable(力变量)
Translational Mechanical Systems (平动机械系统)	***M*** **: inertia** (惯量) **mass** (质量) 	***K*: stiffness** (刚度) **spring** (弹簧) 	***B*: damping** (阻尼) **damper** (阻尼器) 	**Displacement** (位移) ***x*** **[m]**	**Velocity** (速度) ***v*** **[m/sec]**	**Force** (力) ***f*** **[N]**
Rotational Mechanical Systems (旋转机械系统)	***J*** **: inertia**(惯量) **Gear** (齿轮) 	***K*: stiffness** (刚度) **Shaft** (弹性轴)	***B*: damping**(阻尼) **bearings, bushings** (轴承、衬套) 	***ϑ*:Angular displacement** (角位移) **[rad]**	***ω*: Angular velocity** (角速度) **[rad/sec]**	***τ*: torque** (力矩) **[Nm]**
Electrical Systems (电路系统)	***L*: inductance** (电感) **Inductor** (电感器) 	***C*: capacitance** (电容) **capacitor** (电容器) 	***R*: resistance** (电阻) **Resistor** (电阻器) 	***q*: charge** (电荷) **[C]**	***i*: current** (电流) **[A]**	***e*:voltage** (电压) **[V]**

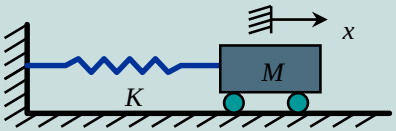
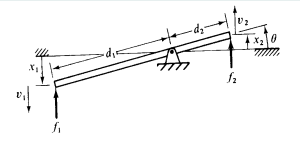
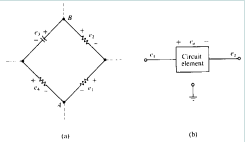
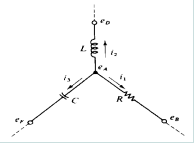
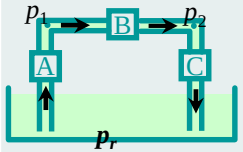
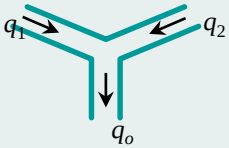
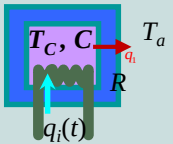
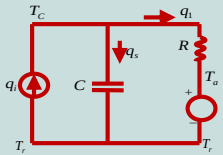
Primary Elements (cont.) (主要元件)

System(系统)	Inertia Element: Store Kinetic Energy (惯性元件：储存动能)	Stiffness Element: Store Potential Energy (刚度元件：储存势能)	Friction Element: Dissipate Energy (摩擦元件：耗散能量)	Displacement variable(位移变量)	Velocity variable(速度变量)	Force variable(力变量)
Fluid Systems (流体系统)	**Fluid inertance (Inductance)(流体惯性(电感)): long pipes(长管道)** 	**Fluid capacitance (流容): cylinder chambers, tanks, accumulators (气缸室、油箱、蓄能器)** 	**Fluid resistance (流阻): orifices, valves, nozzles, friction in pipes (孔口、阀门、喷嘴、管道摩擦)** 	***V*: volume (体积) [m³]**	***q*: volumetric flow rate (体积流量) [m³/sec]**	***p*: pressure (压力) [N/m²]**
Thermal Systems (热系统)	**Null**	**Thermal capacitance (热容): 不同材料的物质具有不同热常数** 	**Thermal Resistance (热容): conduction, convection, radiation (传导、对流、辐射)** 	**Null**	***q*: heat flow rate (热流量) [J/sec = W]**	***T*: temperature (温度) [ºK] or [ºC]**

Basic Modeling Elements（基本建模元件）

	Inertia Element 	Stiffness Element 	Friction Element 	Displacement variable	Velocity variable	Force variable
Translational Mechanical Systems	$M\ddot{x} = \sum_i f_i$	$f_S = K\underbrace{(x_2 - x_1)}_{\delta x}$	$f_D = B(\dot{x}_2 - \dot{x}_1)$	**Displacement** ***x*** **[m]**	**velocity** ***v*** **[m/sec]**	***Fore*** ***f*** **[N]**
Rotational Mechanical Systems	$J\ddot{\theta} = \sum_i \tau_i$	$\tau_S = K(\theta_2 - \theta_1)$	$\tau_D = B(\dot{\theta}_2 - \dot{\theta}_1)$	***ϑ:*****Angular displacement [rad]**	***ω:*** **Angular velocity [rad/sec]**	***τ:*** **torque [Nm]**
Electrical Systems	$e_{12} = e_L = L\left(\frac{d}{dt}i\right)$	$q_c = C\,e_C \Leftrightarrow i = C\left(\frac{d}{dt}e_C\right)$	$e_R = R\,i \Leftrightarrow i = \frac{1}{R}e_R$	q_c**: charge [C]**	***i:*** **current [A]**	***e*****:voltage [V]**
Fluid Systems	$I\frac{d}{dt}q = p_1 - p_2$	$C\frac{d}{dt}\underbrace{(p_c - p_{ref})}_{p_{Cr}}$ $= C\cdot\dot{p}_{Cr} = q_{IN} - q_{OUT}$	$\Delta p = p_1 - p_2 = p_{12} = R\cdot q$ $q = \frac{1}{R}\Delta p = \frac{1}{R}p_{12}$	***V:*** **volume [m³]**	***q:*** **volumetric flow rate [m³/sec]**	***p:*** **pressure [N/m²]**
Thermal Systems	**Null**	$c_p M\frac{d}{dt}T_C = q_{IN} - q_{OUT}$	$q = \frac{1}{R}(T_1 - T_2) = \frac{1}{R}\Delta T$ $q = \frac{\alpha A}{d}\Delta T \Rightarrow R = \frac{d}{\alpha A}$	**Null**	***q:*** **heat flow rate [J/sec = W]**	***T:*** **temperature [ºK] or [ºC]**

Interconnection Relationships （互联关系/物理规律）

	Law1	Law2	Diagram
Translational Mechanical Systems	**Newton's Second Law** (牛顿第二定律) $\frac{d}{dt}(Mv) = M\ddot{x} = \sum_i f_{EXTi}$	**Newton's Third Law** (牛顿第三定律) -Action & Reaction Forces (作用力和反作用力)	
Rotational Mechanical Systems	**Newton's Second Law** (牛顿第二定律) $\frac{d}{dt}\underbrace{(J\omega)}_{\text{Angular Momentum}} = J\ddot{\theta} = \sum_i \tau_{EXTi}$	**Newton's Third Law** (牛顿第三定律) -Action & Reaction Forces (作用力和反作用力)	
Electrical Systems	**Kirchhoff's Voltage Law (loop law)** (基尔霍夫电压定律) $\sum_{\text{Closed Loop}} e_j = 0$	**Kirchhoff's Current Law (node law)** (基尔霍夫电流定律) $\sum_{\text{Any Node}} i_j = 0$	 
Fluid Systems	**Compatibility Law** (相容性定律) $\sum_{\text{closed loop}} \Delta p_j = \sum \Delta p_{ij} = 0$	**Continuity Law** (连续性定律) $\sum_{\text{Any Node}} q_j$ or $\sum q_{IN} = \sum q_{OUT}$	 
Thermal Systems	**Energy Balance - 1st Law of Thermodynamics** (热平衡-热力徐第一定律) $q_{store} = \sum q_{IN} - \sum q_{OUT} + q_{\substack{GENERATED' \\ WITHIN}} + \frac{d}{dt}W_{\substack{WORK \\ DONE}}$		 

Energy Distribution（能量分配）

- **EOM of a Mass-Spring-Damper System**（弹簧–质量–阻尼系统）

$$\underbrace{J\ddot{\theta}}_{\substack{\text{Contribution}\\\text{of Inertia}}} + \underbrace{B\dot{\theta}}_{\substack{\text{Contribution}\\\text{of the Damper}}} + \underbrace{K\theta}_{\substack{\text{Contribution}\\\text{of the Spring}}} = \underbrace{\tau(t)}_{\substack{\text{Total}\\\text{Applied Torque}}}$$

We want to look at the energy distribution of the system.

- **Multiply the Equation by Angular Velocity Term ω**（方程两边乘以角速度）

$$J\ddot{\theta}\cdot\dot{\theta} + B\dot{\theta}\cdot\dot{\theta} + K\theta\cdot\dot{\theta} = \tau(t)\cdot\omega$$

- **Integrate the above Equation w.r.t. Time**（方程两边积分）

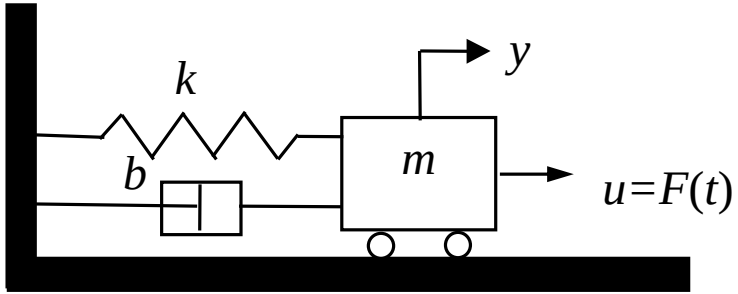
$$\underbrace{\int_{t_0}^{t_1} J\ddot{\theta}\cdot\dot{\theta}\,dt}_{\Delta KE} + \underbrace{\int_{t_0}^{t_1} B\omega\cdot\omega\,dt}_{\int_{t_0}^{t_1} B\omega^2 dt \,\geq\, 0} + \underbrace{\int_{t_0}^{t_1} K\theta\cdot\dot{\theta}\,dt}_{\Delta PE} = \underbrace{\int_{t_0}^{t_1} \tau(t)\cdot\omega\,dt}_{W}$$

ΔKE	$\int_{t_0}^{t_1} B\omega^2 dt \geq 0$	ΔPE	W
$\Downarrow$	$\Downarrow$	$\Downarrow$	$\Downarrow$
Change of kinetic energy	**Energy dissipated by damper**	**Change of potential energy**	**Total work done by the applied torque τ(t) from time t_0 to t_1**

Modeling Steps（建模步骤）

- **Understand System Function, Define Problem, and Identify Input/Output Variables**（理解系统功能，定义问题，识别输入/输出变量）
- **Draw Simplified Schematics Using Basic Elements**（画出采用基本元件的简化原理图）
- **Develop Mathematical Model (Diff. Eq.)**（建立数学模型）
  - Identify reference point and positive direction（识别参考点和正方向）
  - Draw free-body-diagram (FBD) for each basic element（画出每个基本元件的自由体受力分析图）
  - Write elemental equations as well as Interconnecting Equations by applying physical laws. (*Check: # eq = # unk*)（采用物理规律写出元件方程和互联关系方程）
  - Combine equations by eliminating intermediate variables（合并方程以便消除中间变量）
- **Validate Model by Comparing Simulation Results with Physical Measurements**（比较仿真结果和物理测量以验证模型）

Input/Output Models vs State-Space Models



Mass-spring-damping system

Input-Output Model

$$m\ddot{y} + b\dot{y} + ky = u$$

$$\ddot{y} + \frac{b}{m}\dot{y} + \frac{k}{m}y = \frac{1}{m}u$$

Transfer function

$$\frac{Y(s)}{U(s)} = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}}$$

State-Space Model

$$x_1 = y, x_2 = \dot{y}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}}_{A(t)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} x + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{B(t)} u$$

$$y = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{C(t)} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} x + \underbrace{[0]}_{D(t)} u$$

Input/Output Models vs State-Space Models

	State-Space Models	Input / Output Models
Nonlinear Systems	$\dot{x} = f(t, x, u)$ $y = h(t, x, u)$	$l\left(y^{(n)}(t), y^{(n-1)}(t), \cdots, y(t),\ u^{(n-1)}(t), \cdots, u(t), t\right) = 0$
Linear Systems	$\dot{x} = A(t)x + B(t)u$ $y = C(t)x + D(t)u$	$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_0 y(t) = b_{n-1}u^{(n-1)} + \cdots + b_0 u(t)$
Features	• Consider the internal behavior of a system（考虑系统的内部行为） • Can handle complicated output variables easily（容易处理复杂输出变量） • Have significant computation advantage for computer simulation（有利于计算机仿真） • Can represent multi-inputs-multi-outputs (MIMO) systems and nonlinear systems（能表示多输入多输出系统和非线性系统）	• Conceptually simple（概念简单） • Easy to be converted to frequency domain transfer functions that are more intuitive to practicing engineers（容易转化为实践工程师更直观面对的频域传递函数） • Difficult to solve in time domain (solution: using Laplace transformation and solve in frequency domain)（较难解决时域响应(采用拉氏变换或在频域求解)）

Linearization（线性化）

- **Nonlinear System Model**

$$\dot{x} = f(x,u)$$

$$y = h(x,u)$$

- **Equilibrium Points** **or** **Equilibrium Trajectory**

$$0 = f(x_e,u_e) \quad (x_e, y_e, u_e)$$

$$y_e = h(x_e,u_e)$$

$$\dot{x}_e = f(x_e(t),u_e(t)) \quad (x_e(t), y_e(t), u_e(t))$$

$$y_e(t) = h(x_e(t),u_e(t))$$

- **Taylor Series Expansion at EquilibriumTrajectory/Point**（泰勒展开和平衡轨迹/点）

$$\delta x = x(t) - x_e(t), \delta y = y(t) - y_e(t), \delta u = u(t) - u_e(t)$$

$$\delta\dot{x} = \dot{x} - \dot{x}_e = f(x,u) - f(x_e,u_e) = \left.\frac{\partial f}{\partial x}\right|_{\substack{x=x_e\\u=u_e}} \delta x + \left.\frac{\partial f}{\partial u}\right|_{\substack{x=x_e\\u=u_e}} \delta u + H.O.T$$

$$\delta y = y - y_e = h(x,u) - h(x_e,u_e) = \left.\frac{\partial h}{\partial x}\right|_{\substack{x=x_e\\u=u_e}} \delta x + \left.\frac{\partial h}{\partial u}\right|_{\substack{x=x_e\\u=u_e}} \delta u + H.O.T$$

Linearization

- **Small Perturbation and Linear Approximation**

$$\delta\dot{x} = A(t)\delta x + B(t)\delta u$$

$$\delta y = C(t)\delta x + D(t)\delta u$$

$$\delta\dot{x} = \underbrace{\left.\frac{\partial f}{\partial x}\right|_{\substack{x=x_e\\u=u_e}}}_{A(t)}\delta x + \underbrace{\left.\frac{\partial f}{\partial u}\right|_{\substack{x=x_e\\u=u_e}}}_{B(t)}\delta u$$

$$\delta y = \underbrace{\left.\frac{\partial h}{\partial x}\right|_{\substack{x=x_e\\u=u_e}}}_{C(t)}\delta x + \underbrace{\left.\frac{\partial h}{\partial u}\right|_{\substack{x=x_e\\u=u_e}}}_{D(t)}\delta u$$

$$A(t) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_2}{\partial x_1} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \qquad B(t) = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \frac{\partial f_2}{\partial u_1} & \frac{\partial f_2}{\partial u_2} & \cdots & \frac{\partial f_2}{\partial u_m} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial f_n}{\partial u_1} & \frac{\partial f_n}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}$$

$$C(t) = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} & \frac{\partial h_1}{\partial x_2} & \cdots & \frac{\partial h_1}{\partial x_n} \\ \frac{\partial h_2}{\partial x_1} & \frac{\partial h_2}{\partial x_2} & \cdots & \frac{\partial h_2}{\partial x_n} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial h_p}{\partial x_1} & \frac{\partial h_p}{\partial x_2} & \cdots & \frac{\partial h_p}{\partial x_n} \end{bmatrix} \qquad D(t) = \begin{bmatrix} \frac{\partial h_1}{\partial u_1} & \frac{\partial h_1}{\partial u_2} & \cdots & \frac{\partial h_1}{\partial u_m} \\ \frac{\partial h_2}{\partial u_1} & \frac{\partial h_2}{\partial u_2} & \cdots & \frac{\partial h_2}{\partial u_m} \\ \cdots & \cdots & \cdots & \cdots \\ \frac{\partial h_p}{\partial u_1} & \frac{\partial h_p}{\partial u_2} & \cdots & \frac{\partial h_p}{\partial u_m} \end{bmatrix}$$

Example of Linearization

- **viscose friction motion**

Problem:
- planar moving particle (2-dimension)
- viscose friction
- range measurement

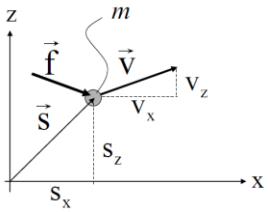
System model:

- dynamics

$$m\dot{\vec{v}} = \vec{f} - c|\vec{v}|^2 \frac{\vec{v}}{|\vec{v}|}$$

- measurement

$$r = |\vec{s}| = \sqrt{s_x^2 + s_y^2}$$



Nonlinear state space model:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{c}{m} x_2 \sqrt{x_2^2 + x_4^2} + \frac{1}{m} u_1$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -\frac{c}{m} x_4 \sqrt{x_2^2 + x_4^2} + \frac{1}{m} u_2$$

$$y = \sqrt{x_1^2 + x_3^2}$$

Linearized state space model:

$$\Delta\dot{x}_1 = \Delta x_2$$

$$\Delta\dot{x}_2 = -\frac{c}{m} \frac{2x_{2*}^2 + x_{4*}^2}{\sqrt{x_{2*}^2 + x_{4*}^2}} \Delta x_2 - \frac{c}{m} \frac{x_{2*} x_{4*}}{\sqrt{x_{2*}^2 + x_{4*}^2}} \Delta x_4 + \frac{1}{m} \Delta u_1$$

$$\Delta\dot{x}_3 = \Delta x_4$$

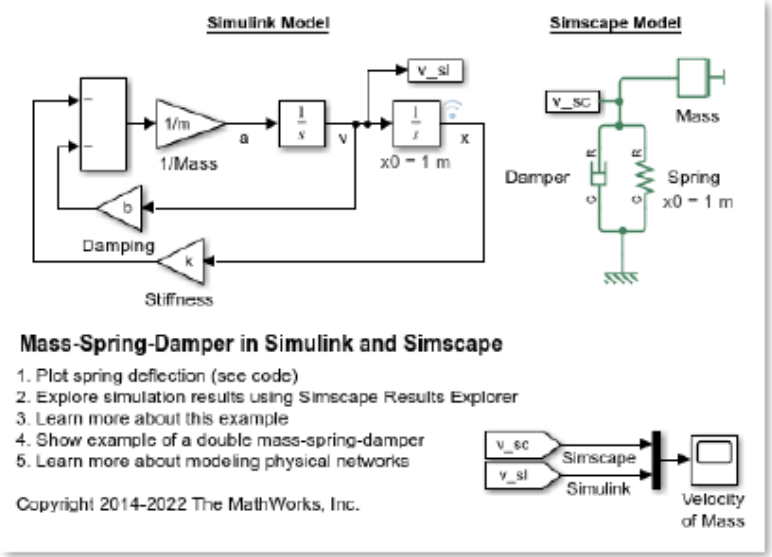
$$\Delta\dot{x}_4 = -\frac{c}{m} \frac{x_{2*} x_{4*}}{\sqrt{x_{2*}^2 + x_{4*}^2}} \Delta x_2 - \frac{c}{m} \frac{2x_{4*}^2 + x_{2*}^2}{\sqrt{x_{2*}^2 + x_{4*}^2}} \Delta x_4 + \frac{1}{m} \Delta u_2$$

$$\Delta y = \frac{x_{1*}}{\sqrt{x_{1*}^2 + x_{3*}^2}} \Delta x_1 + \frac{x_{3*}}{\sqrt{x_{1*}^2 + x_{3*}^2}} \Delta x_3$$

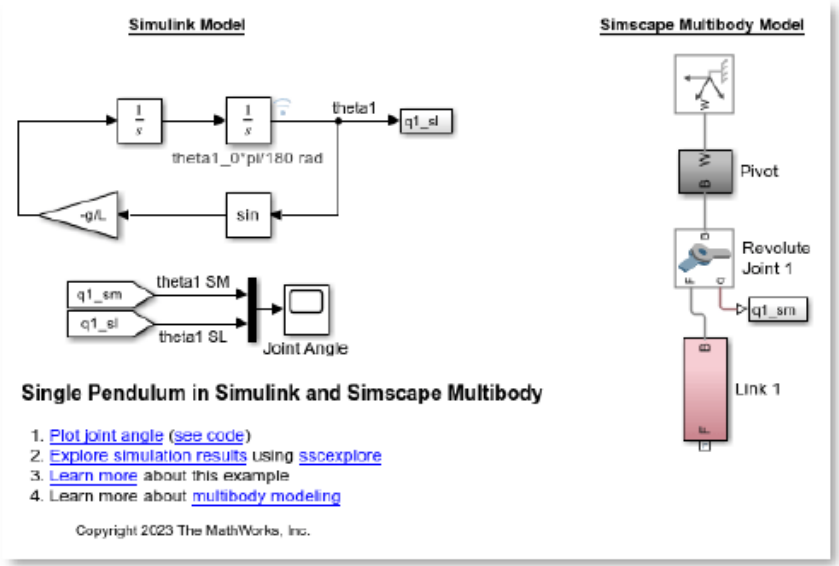
```
clear; close all; clc
%%%
```

Cases in MATLAB

- **Mechanical System**



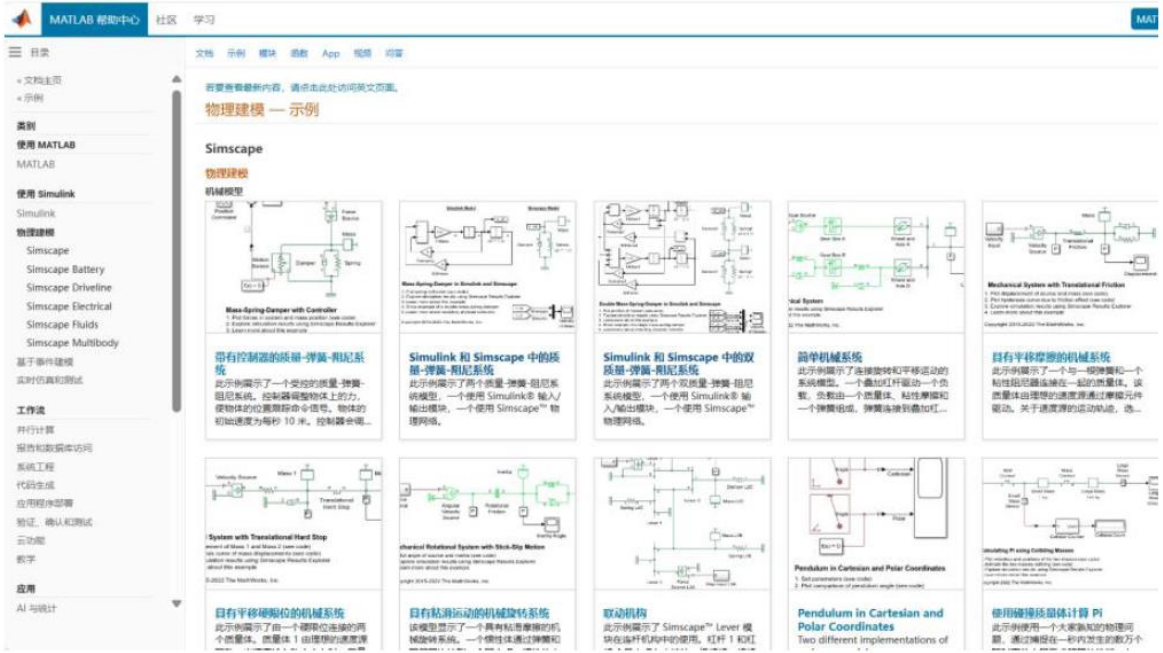
Simulink 和 Simscape 中的质量-弹簧-阻尼系统
https://ww2.mathworks.cn/help/simscape/ug/mass-spring-damper-in-simulink-and-simscape.html

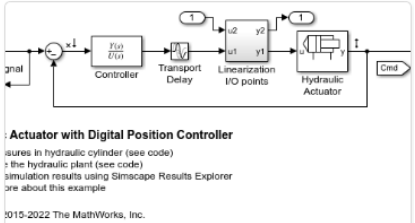


https://ww2.mathworks.cn/help/sm/ug/single-pendulum-in-simulink-and-simscape-multibody.html?s_tid=srchtitle_site_search_1_simscape+simulink

Cases in MATLAB

物理建模 — 示例





带数字位置控制器的液压作动器

此示例展示了与在等温液体域中实现的双作用气缸一起在闭环回路中起作用的二通阀。此控制器表示为连续时间传递函数加传输延迟。当以离散...

https://ww2.mathworks.cn/help/overview/examples.html?category=physical-modeling&s_tid=CRUX_topnav

https://ww2.mathworks.cn/help/simscape/mechanical-systems.html?category=mechanical-systems&s_tid=CRUX_topnav